Marin mid-training and post-training: concrete experiments

Long-context data stores, matched continuation runs, chat SFT and RL probes

Track the experiment, not just a stage name

Marin uses several model families and experiment branches. The older dense Marin-32B cooldown, the June 67B-A2B Grug context-extension/SFT work, Delphi dense runs and later Snowball RL experiments are not one uninterrupted public recipe. This guide follows individual code/data/checkpoint lineages. The data build guide and evaluation guide provide their input and measurement contracts.

All PR merge states below were retrieved on **2026-11-04**. The evidence bundle saves metadata, selected head files, experiment-issue discussions and HF cards. A PR head records reviewable source; an issue's **runtime source SHA** records what its author says was launched. They need not match. No continuation, SFT or RL run has been executed here.

1. Long-context mid-training begins with a different data inventory

PR #8637, unmerged, head `c2fb62b759c3257e4fcb673922431f76ee84acfc`, builds a **separate length-partitioned store**, rather than rewriting the canonical source DAG. Its saved builder partitions retained documents within domain × quality cells at **65,536 tokenizer tokens**.

Author-reported output	Short: ≤65,536 tokens	Long: >65,536 tokens
Documents	11,722,087,779	6,470,523
Tokens	8,554,200,622,073	1,818,143,081,980
Populated cells	200	191

The PR reports 114 source components, 391 populated child cells and per-cell count conservation. The store URI is `gs://marin-us-central2/datakit/store/june-67b-a2b-length64k/2026.08.24`. The **counts are reported in the PR**, not obtained by our own scan, and the URI's existence in code is not proof of public read access. Long-context training cannot be reproduced from the public mixture-results parquet alone.

The intervention preserves outer topic/quality weights. Let w_b be the original weight of cell b, $U_{b,S}$ and $U_{b,L}$ its available short/long tokens, and s the long-document skew. Then

$$w_{b,L} = w_b \frac{sU_{b,L}}{U_{b,S} + sU_{b,L}}, \qquad w_{b,S} = w_b \frac{U_{b,S}}{U_{b,S} + sU_{b,L}}.$$

Thus $w_{b,L} + w_{b,S} = w_b$. The aggregate long share is $\sum_b w_{b,L}$, **not** the corpus-wide long-token fraction. The reported weighted shares are about **14.943% at 1×**, **22.586% at 2×**, and **43.405% at 8×**. Increasing s by eight does not multiply the final global long share by eight. PR #8637, 8× experiment #8813.

The sampler changes the realized treatment

A review of PR #8656 identified a naive flattened-child implementation: flooring weights on a 49,152-slot mixture lattice zeroed 79 positive cells and changed nominal long-token share from 22.5856% to 22.4365%, assigning the leftover slots to a large short cell. The retrieved head's mixture code instead composes short/long children with repetition inside the fixed

outer cells. The review is evidence of a **historical issue and revised implementation**, not a claim that the retrieved head still uses the criticized flattening.

Replication gate: inspect realized sampler frequencies, zero-probability cells, child lengths and complete-cycle token counts. A mathematically normalized vector is insufficient if discrete sampling changes it. For cyber logs/repositories, preserve document boundaries and measure actual document-length supply before applying this treatment.

2. Matched 65K→262K continuation runs

PR #8656, unmerged, head `03890a0c6497075ae570bf7dc57fe6aa5419e967`, provides pinned context-extension launchers, with later 4× PR #8791 and 8× PR #8815, also unmerged. The treatment is a data change from the same continuation checkpoint, rather than training unrelated models from scratch.

The 8× experiment contract records:

Controlled quantity	Configuration
Initial model	June 67B-A2B Grug cooldown, native `step-156000` checkpoint
Continuation	1,000 updates, ending at step 157000; context 65,536→262,144
Global sequence batch	256; **67,108,864 tokens/update**
Total continuation exposure	**67,108,864,000 tokens**, calculated as batch × context × updates
Attention setting	QK multiplier 1.75
Optimizer/schedule	MuonH; linear LR from 0.000264 to 0.0000528
Mesh	replica 1, data 256, context 4, expert 1, model 1
Data offset	`resume_data_offset=7,296,000`
Evaluation/permanent checkpoint interval	Every 250 updates
Experimental axis	Only the within-cell long-document repetition/skew

Its initial checkpoint URI is recorded in the issue; the manifest describes a ~1.079TB native checkpoint. This is a TPU v4-2048 experiment contract, not a directly runnable 96-H100 configuration. Reproduce the scientific controls on qualified hardware, or budget and qualify the exact stack separately.

What was launched, and what completed?

Treatment	PR / experiment	Runtime SHA reported in discussion	Evidence retrieved
2×	#8656 / #8734	`1691d0a9afbfe849538e180186bec7759574cf2c`	Launch, scheduler trouble/recovery, then completion at step 157000 with final checkpoint metadata reported present
4×	#8791 / #8790	`3c82e92a3d4c0d6b286038811dc7c330a2253017`	Completion at step 157000 with final checkpoint metadata reported present
8×	#8815 / #8813	`92514c5358e9cd99c89bdd4a6600ac7bbeddc147`	Launch verified, workers allocated and W&B initialized; no final scored comparison established by the retrieved comments

A closed experiment issue or successful launch is not a quantitative result. We have not retrieved a complete paired, per-length MRCR/RULER result table establishing which skew won. The supplied evidence justifies replicating this **controlled experiment**, not choosing 8× as empirically optimal. For the 2× completion, checkpoint presence is author-reported; our collector did not access that cloud object.

The operational record also matters: the 2× job initially could not form its worker gang because the slice had inconsistent worker IDs. Record unstarted time separately from training throughput. Otherwise wall-clock comparisons can penalize a data recipe for infrastructure trouble.

3. A separate 4K→16K hero handoff

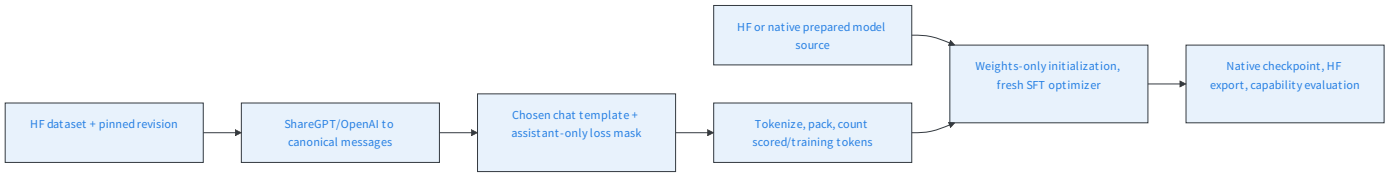
PR #9639, unmerged, head `7e1a48c4e6544c9f60adaabc8ab63b87a1f3948b`, addresses a different hero run. It holds **4,194,304 tokens per rack per update** constant as context changes; at 16K on 11 racks that becomes 2,816 sequences. Step count remains 390,251 with LR/decay/router settings preserved.

Its most consequential data detail is shuffling fixed 2^{20}**-token blocks**, rather than fixed sequence-count blocks. The PR estimates that retaining the old sequence-based convention after the switch would reread **2.85T tokens**. The main-mixture boundary shifts from step108000 to108096 and requires explicit acceptance. Canonical evaluation stays at 4K, with separate longer-context diagnostics and cumulative-FLOP continuity.

The PR explicitly says the live run is unchanged and a GPU rehearsal is still required. PR #10711 later proposes multiple validation contexts; its verification is CPU-only. For our project, the actionable checklist is: restore optimizer and sampler state, preserve token-level exposure accounting, check actual shuffle continuity, record context/QK/RoPE changes and evaluate at both the old fixed context and the new context. A context-length flag alone does not specify a sound continuation.

4. Chat SFT: data preparation and weights-only initialization

PR #7231, merged, head `16e86616bc49551d9f194190f94eb33d9a7c38db`, introduces `SFTSpec` and an ArtifactStep pipeline:



A generic smoke in that PR reached train-step tracing but was **not fully green** due to tied-embedding µP optimizer-state vocabulary padding. Merge status is not a completed end-to-end smoke. PR #7280, merged, prepares Delphi's reserved think/tool token rows and corresponding embeddings/head before SFT.

Concrete two-stage MoE recipe

PR #7250, merged, head `9f527141435cc3606511b331326d09b1f92d696b`, adapts the generic interface to Grug's native model tree. The baseline source records:

Stage	Input / revision in code	Initialization / horizon
Plain chat	`nyu-dice-lab/wildchat50m-rewild-sft-385700`, `46a5bb5`	Weights-only from the native step42150 base; one packed epoch
Science reasoning	`laion/llama-nemotron-science-reasoning-on-canonical-think-full`, `bae881d`	Stage1 output; one packed epoch

These are short revision pins in the source; resolve and save full HF commits before our launch. Canonicalize conversation columns, preserve assistant-only masks and chain actual Stage1 output into Stage2. Packed epoch count comes from the resulting token cache, not the unprocessed number of conversations. The code's ~538.9M WildChat-token estimate gives about257 steps at sequence32768/batch64; it is a configuration comment, not our measured token census.

The implementation uses **64 H100 GPUs**, EP8, sequence32768, global batch64 and model-axis1. It uses **AdamH**, rather than blindly retaining the base run's Muon configuration: the PR explains a replicated ~21GiB Newton–Schulz workspace causing OOM at the considered geometry. The code sets both matrix and auxiliary Adam LRs to `5e-5`, betas `(0.9,0.95)`, epsilon `1e-`

`8`, clipping1, weight decay0, warmup3%, cosine schedule and minimum ratio0.1. These are **first-pass recipe values**, not a demonstrated hyperparameter optimum.

The PR reports a completed **eight-step smoke**, loss0.712, checkpoint saved and ~68GiB/device memory. That verifies a particular systems path, not final instruction-following or reasoning quality. Run the evaluation suite after training; do not promote the smoke loss into a capability result. PR #7250.

Dense Delphi is a separate recipe

The Delphi SFT config uses a Magpie/science warmup mixture and a prepared HF model, with its own optimizer/schedule. We saved the exact base HF revision `56cc5aaf65cd59d6632962cee636a0423abfbf66`. Some dataset refs in that config remain `main`/TODO and must be pinned. Do not substitute this recipe into the native MoE launcher without testing the model source, tokenizer and chat template.

5. RL probes and synthetic curriculum evaluation

PR #6325, closed **unmerged**, head `4c2ac5da0d5e5137a88d206b647afd6b348a6204`, defines a 100-step MATH500 RL probe over a grid of mid-trained Delphi checkpoints: three mixtures × nine budgets, plus reference cold starts. It specifies RLOO/math-answer formatting. The launcher is useful evidence of the proposed comparison design; it does not establish that all those RL runs completed. We pinned and saved one public Magpie cold-start model's card/config, without downloading weights.

PR #10623, unmerged, head `cb5a332a5b7f137fcee0eaccbfc36503a0037328`, gives a more developed evaluation account for recursive synthetic improvement. It compares a pool-only control with a pool-plus-synthesized treatment, evaluates held-out pool tasks and probes of synthesized/held-out families, and reports paired bootstrap intervals over tasks with four attempts per task.

The Snowball example's synthesized-family probe gain coexists with a Smartie validation decline; Qwen seed1's held-out-family probe gain coexists with an approximately unchanged Smartie score. This is concrete evidence that specializing training can improve a targeted distribution without improving general held-out tasks. The evaluation guide lists exact numbers and limits. Our saved README includes additional seeds and caveats; we have not independently re-scored its per-attempt traces.

For our cyber extension, keep source-document adaptation, chat SFT and verifier-driven agent RL as separate experiments. First validate representations/formatting and export parity, then compare equal-budget base/control/treatment policies on independent tasks and general guardrails. A cyber verifier must record environment failures separately from incorrect behavior. Those are proposed replication controls, not results already established for our intended model.

6. What to reproduce next

1. **Data:** build and inspect one versioned short/long store with conserved cell counts and known realized sampler shares.
2. **Continuation:** qualify a shorter-context checkpoint restore, optimizer/sampler continuity and native/exported loss before attempting262K sequences.
3. **Evaluation:** freeze fixed-context BPB, long-context likelihood and generated-answer retrieval as separate cohorts; archive per-item scores and coverage.
4. **SFT:** reproduce a small canonical-message → packed-cache → assistant-mask → weights-only SFT → export path; expand to the chosen native model only after it passes.
5. **RL:** reproduce one verified task/environment/control comparison and its uncertainty calculation before scaling the curriculum.

These steps are an executable research order once datasets/checkpoints and the intended hardware environment are available. The branch-specific source and reported runs tell us exactly what to inspect; access to their private or unverified cloud artifacts remains an explicit dependency.